NeuroText

EEG representation learning for sentence retrieval

Can EEG retrieve sentence meaning beyond matched negative controls?

ZuCo 1.0 + 2.0 | Statistical ridge · CNN · SSL · frozen LaBraM

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Executed within-dataset results; strict cross-dataset evaluation remains pending.

Define the prediction task

Fixed-pool retrieval with unseen readers

1. Encode a sentence-length EEG recording.
2. Predict its location in a frozen sentence-embedding space.
3. Rank unique candidate sentences by cosine similarity.
4. Evaluate Recall@1/5/10 and mean reciprocal rank (MRR).

Above random ranking alone is insufficient: real EEG must beat matched controls.

Data preparation and quality audit

Normal reading · 105 ordered channels · resampled to 200 Hz

ZuCo 1.0

12 readers

3,356 prepared observations

3,349 after duration QC

ZuCo 2.0

18 readers

5,911 prepared observations

5,884 after duration QC

19-channel LaBraM adaptation is a separate cohort: 3,345 / 5,838 observations.

One shared, memory-conscious implementation

Windows + Colab clients use the same Python package

LOAD

Lazy signal arrays

Audited manifests

REPRESENT

Statistics / CNN

SSL / frozen LaBraM

ALIGN

Train-only scaling

Validation selection

RANK

MiniLM targets

Cosine retrieval

Local hardware: RTX 3050 with 4 GB VRAM; lazy loading limits memory use.

Shared code and documented inputs make weak results inspectable and reproducible.

Controls make the central comparison

Hold out subjects; fit preprocessing on training observations only

Real EEG: original signals and true sentence pairing.

Gaussian: matched per-channel scale, independently generated noise.

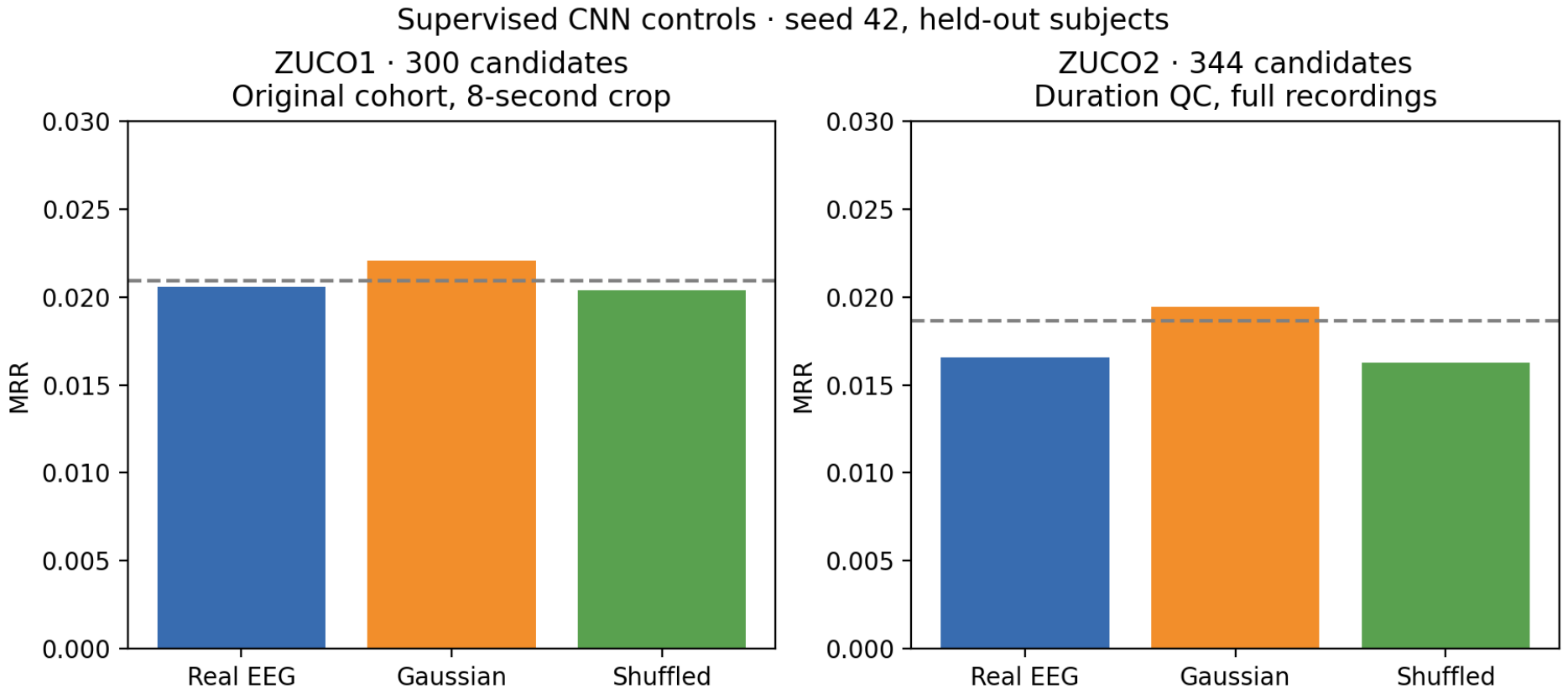
Shuffled: separately trained model with incorrect sentence pairing.

Compare within the same cohort, split, seed and candidate pool.

Single-split differences are descriptive; no statistical significance is claimed.

Supervised CNN: Gaussian matches top-1 hits

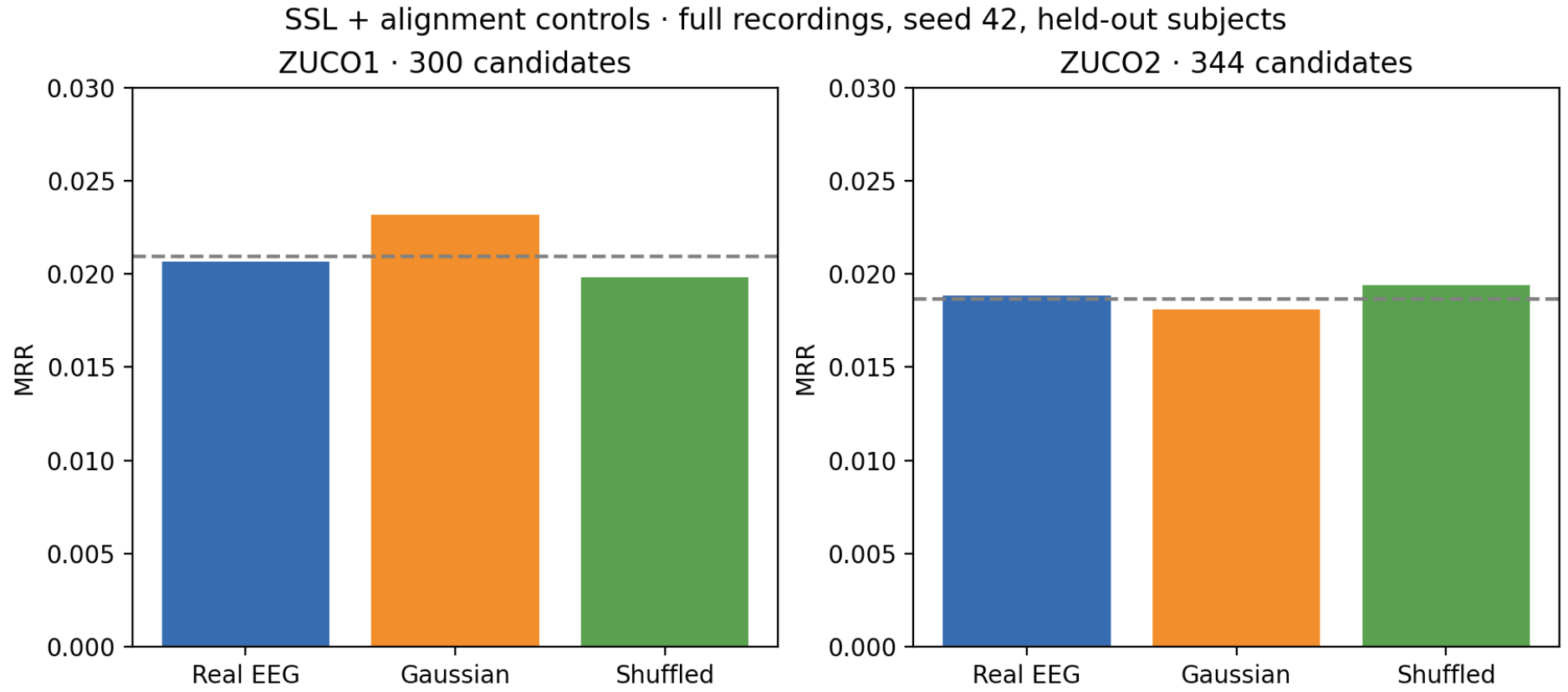
Seed 42 · ZuCo1: 592 queries / 300 candidates · ZuCo2: 984 / 344



Gaussian MRR exceeds real EEG in both datasets.

SSL: reconstruction does not establish decoding

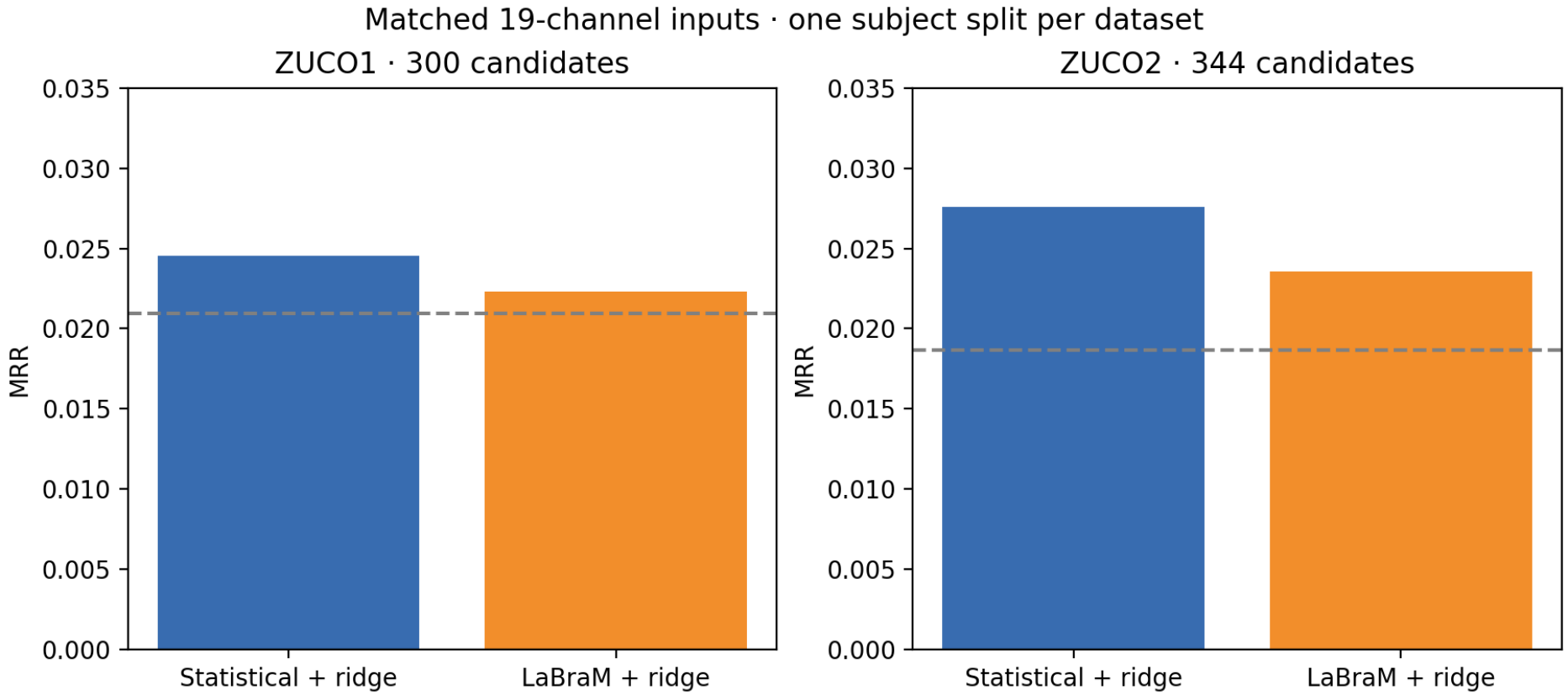
Full recordings · duration QC · Gaussian control covers both learned stages



No consistent retrieval advantage for real EEG over the controls.

Frozen LaBraM against matched statistical features

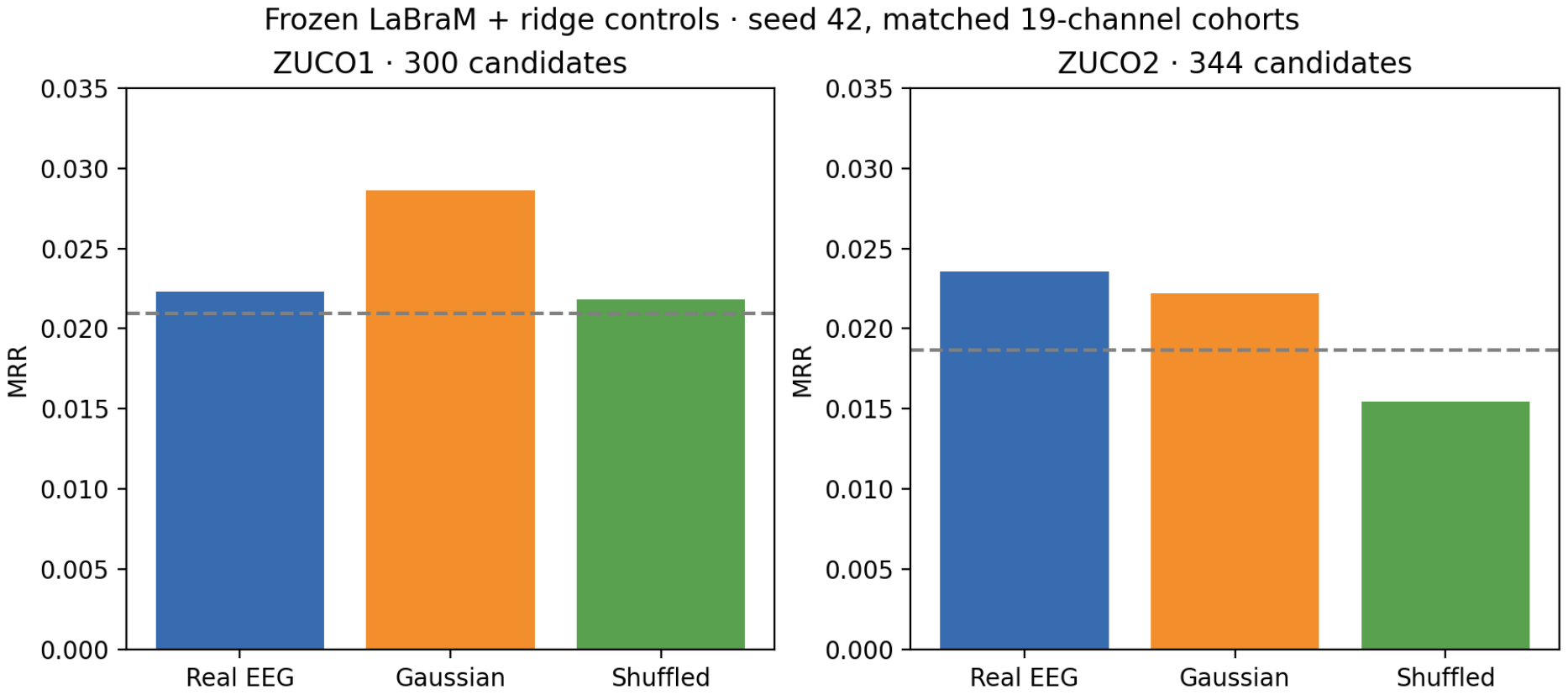
Identical adapted signals · 19 mapped electrodes · whole-second support



Statistical features have higher top-1 and MRR on both reported splits.

LaBraM controls limit the interpretation

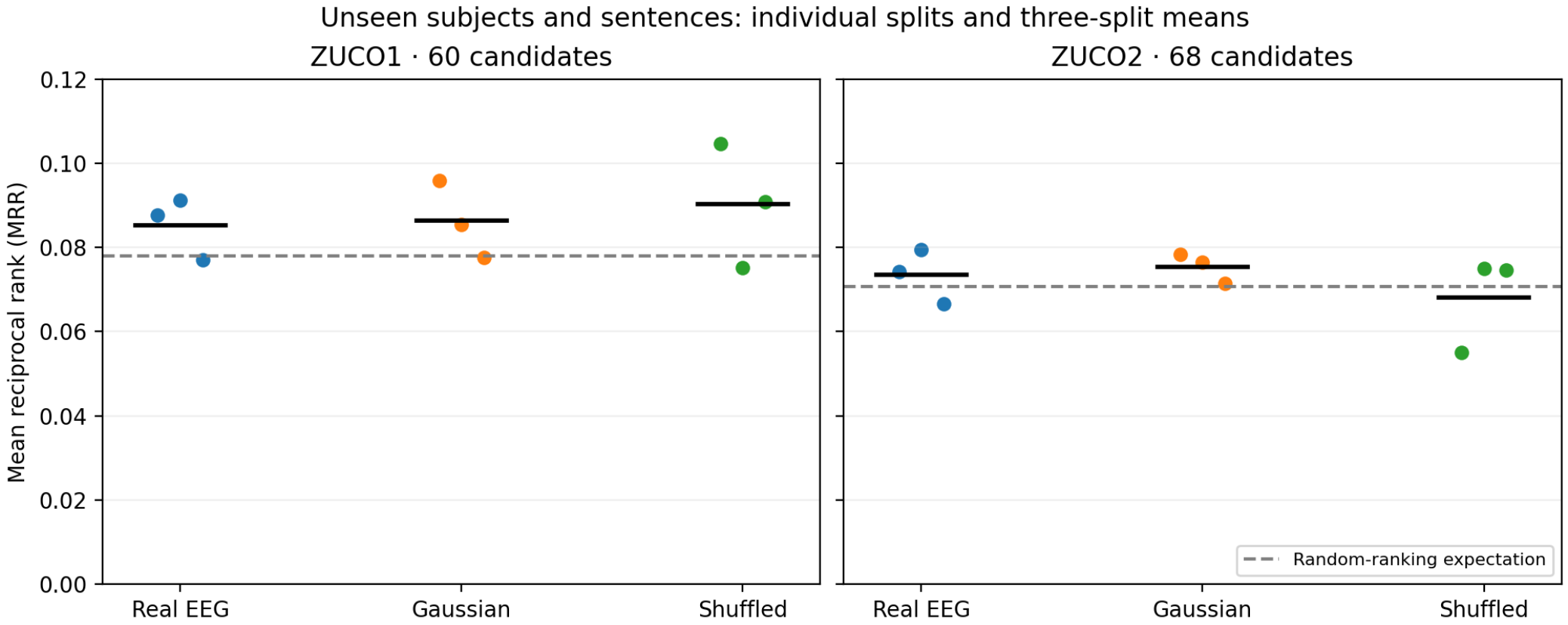
Frozen backbone · separately fit ridge models · same adapted cohort and split



No consistent real-EEG superiority under this exploratory adaptation.

Unseen subjects and unseen sentence content

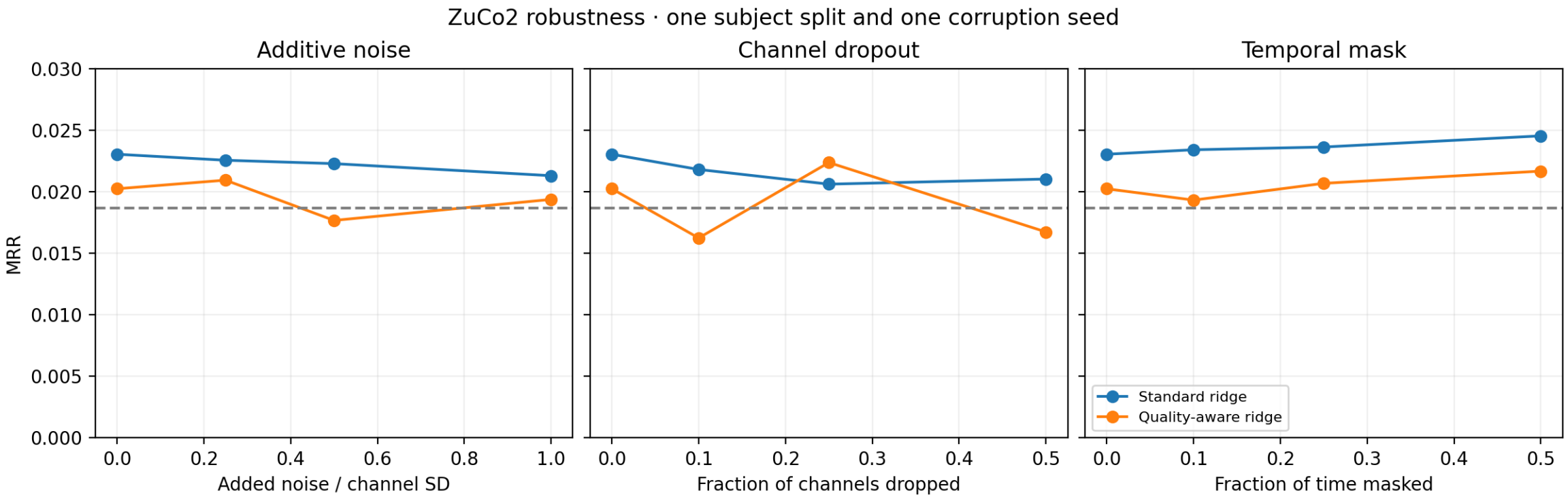
Ridge · seeds 42, 43, 44 · 60 candidates in ZuCo1 / 68 in ZuCo2



Real EEG exceeds Gaussian MRR on only one of three seeds in each dataset.

Quality weighting shows no consistent benefit

ZuCo2 ridge · 984 queries / 344 candidates · one split and corruption seed



Quality-aware MRR is lower in 8 of 9 nonzero corruption conditions.

Published work: compare protocols before scores

Generation, classification and retrieval measure different tasks

EEG-to-text papers often report BLEU / ROUGE for language generation.

LaBraM originally reports classification tasks.

Our task reports Recall@K / MRR for a fixed sentence pool.

The original AAAI 2022 repository documents an evaluation correction.

Primary-source protocol review is complete; competitive numerical ranking is unsupported.

Cross-dataset transfer: prepared, not executed

Both directions remain pending participant identity evidence

Prepared: 9,233 signal references and 594 shared sentence targets.

Directions: ZuCo1 → ZuCo2 and ZuCo2 → ZuCo1.

Pending: verify whether the same people appear in both releases.

Then: reconcile identities and fit / select models on the source only.

Preparation is successful; it is not an executed transfer result.

Conclusion and remaining work

A reproducible negative result with explicit limits

Finding: tested representations do not show reliable retrieval beyond controls.

Contribution: modular pipeline, audited data, matched controls and clear limits.

Remaining: identity evidence, strict transfer experiments and final review.

Archive: original evaluation JSON, configs, splits and checkpoints.

These findings do not establish that EEG contains no semantic information.